

SOFT-AG: Soft Robotics for Transforming Agriculture through Adaptive Intelligence in Specialty Crops

I. Research and Capacity-Building Goals and Vision

A. Overview

The global population is aging at an unprecedented rate. By 2050, the number of individuals aged 60 and older is expected to double from 2017 levels, reaching 2.1 billion, nearly 22% of the projected 9.7 billion global population, according to the United Nations. This demographic trend has direct implications for agriculture: the demand for food will rise, yet the labor pool—especially among growers—will shrink as aging populations exit the workforce. In the United States, this challenge is particularly acute in rural and agricultural regions, where the average age of farmers has already surpassed 57 years. West Virginia (WV) exemplifies this national trend. With one of the oldest populations in the country and a predominantly rural landscape, the state faces critical labor shortages in agriculture and an urgent need for technological intervention. To sustain food production and meet both domestic and global demand, U.S. growers—including those in WV—must adopt integrated, smart farming systems that increase productivity while reducing reliance on manual labor. **These efforts will require robust innovation in agricultural robotics, automation, and decision-support technologies tailored** Robotics has transformed modern-day agriculture [1], yet there exists significant unrealized potential for agriculture in rural states such as WV. In fact, **WV ranks 44th in agricultural output amongst the 50 states** [2]. WV's mountainous terrain and smaller farm sizes limit large-scale crop production; however, it has great potential for nice specialty products to offer additional locally produced fresh food options in the region. WV's academic research and development (R&D) enterprise is well-positioned to address the scientific challenges and opportunities at the intersection of soft robotics, agricultural technology, and artificial intelligence. However, to fully capitalize on this potential, coordinated investment in building capacity and infrastructure are essential to overcome existing barriers.

This proposal will address this potential through the development of transformative soft robotics technologies in a team of four major institutions in the state. West Virginia University (WVU), a land-grant flagship of the state, will lead this initiative as the state's sole Research-1 university. The team will also include Marshall University (MU), WV's second largest research university (R-2); West Virginia State University (WVSU), which is the state's other land-grant and HBCU; and Fairmont State University (FSU), a Primarily Undergraduate Institution (PUI) whose student body is composed of over 90% native West Virginians that provide transformational educational opportunities.

Over the past decade, WVU has developed a focus on robotics research and education, leading several high-profile competitive research grants, leading students to secure titles in numerous national and international (over 20 top-ten finishes and 9 first place finishes in national or international student robotics competitions) robotics engineering competitions, and being one of the small number of R1 engineering schools in the country to offer a formal undergraduate degree program in robotics engineering. As an R1 land-grant, WVU is a leader on the state's agriculture research and development and statewide agriculture extension. This mission is strengthened by WVSU. MU's engineering focuses on mechanical and engineering and computer science complements those of WVU. Finally, FSU is an established leader in the state with respect to educational STEM outreach programming and networks, specifically robotics.

This proposal is well aligned with the core mission of the proposed institutions and with the state's overall Science and Technology (S&T) plan. As outlined in the attached Vision 2030: WV S&T Plan, in the past 5 years, WV's research and development grew to over \$300M and the number of STEM PhDs produced grew by 25%. However, WV STEM undergraduate, master's, and overall college-going rates have declined. As such, fundamental research in support of STEM education is a major underpinning mission of the plan as it is strongly motivated with the projected increase in STEM vs non-STEM job prospects given the continued adoption of robotics and automation. SOFT-AG directly aligns with 4 of the 5 research focus areas identified in the plan. First, the application of this proposal falls within the overarching high-priority area of **Life Sciences in Agriculture**, which is one of three sub-areas of focus. Next, directly related to the core research proposed, both **Artificial Intelligence/ Machine Learning** and Advanced Manufacturing, with a specific sub-topic being **Robotics and Autonomy**. SOFT-AG will directly support **STEM education mission**, including K–12 and Higher Education capacity building.

Despite these strengths, WV's research enterprise faces several barriers that we aim to address. First, there is currently **limited faculty density** in emerging areas like soft robotics and AI for agriculture, which constrains cross-disciplinary growth. Fostering this capacity through a major research project will be

imperative. Second, we currently have **insufficient field-deployable test infrastructure** for autonomous systems in agriculture, particularly for specialty crops. This too will be addressed.

B. Vision

The overarching vision of this project is to establish WV as a national leader in soft robotics for sustainable and intelligent specialty crop agriculture. We aim to develop a robust statewide ecosystem for research, education, and innovation that leverages the unique agricultural landscape and rural infrastructure of the Appalachian region. The project's integrated focus on embodied intelligence, adaptive control, and digital twin-driven crop modeling will not only advance the scientific frontier in soft robotics but also ensure translational impact through direct engagement with farmers and local communities.

Long-Term Research Vision (7-Year Horizon)

The long-term vision is to drive foundational advances in robotics, artificial intelligence, and plant-robot interaction while simultaneously strengthening institutional capacity, developing talent pipelines, and enabling translational pathways from lab discoveries to field deployment. Our goal is to design robotic platforms that are responsive to crop and environmental variability, cost-effective for small to mid-sized growers, and capable of empowering farmers by placing adaptable, intelligent robotic tools directly in their hands.

Short-Term Research Goals (First Four Years)

C. Research

In recent decades, robotics has emerged as a transformative force in agriculture, helping to address labor shortages in developed countries and promising to enhance food security. Robots are now capable of detecting diseases, performing phenotyping, and harvesting fruits and vegetables (1). However, these systems face several tangible challenges: 1) Most existing agricultural robots remain confined to laboratory environments due to a lack of robustness, and efficiency required for real-world field deployment. 2) These robots are typically slow and cannot compete with human capabilities. 3) They have limited scalability and adaptability in unstructured outdoor environments. 4) Concerns about safety and crop damage have hindered the integration of robots into farmers' daily lives. Additionally, these systems have become too expensive—both technologically and computationally—relative to human labor, making them inefficient and inaccessible. For example, a human worker can typically identify and harvest some fruit in under **one second** (2), including decision-making. In contrast, many robotic systems require at least **30 seconds** to several **minutes** to complete the same task, with a success rate between **39%** and **78%** (3). As a result, adoption remains limited. To fill this gap, we must think beyond current technology and scale.

Looking deeper into why agricultural robots lag human capabilities reveal critical limitations that contribute to delays in real-world deployment. **One major reason robots remain largely confined to laboratory settings is the inherently dynamic in nature of agriculture.** There is significant variability in crops, which constantly change in color, size, and shape—both at the produce and plant levels—along with fluctuating environmental conditions. **This variability makes robots slower, less adaptable,** and more prone to damaging crops, whereas humans can make decisions quickly and adapt easily (cite).

Another key limitation is the lack of dexterity and limited sensing capabilities that prevent robots from fully understanding their target crops and environments. Integrating all these capabilities into a single system is computationally expensive. Most current agricultural robotic platforms are built using rigid materials, which inherently restrict dexterous movement. This rigidity makes it difficult to embed sensors effectively, leading to occlusions and physical interference that complicate control. As a result, these robots are less agile and struggle to generalize across the unpredictable nature of agricultural environments. They are often unsuitable for handling delicate crops or close interaction with human workers. Although robotic systems can be precise, biology suggests that it is not precision alone—but rather agility, dexterity, and embodied intelligence, combined with multifunctional and multimodal sensing—that makes for effective interaction with the natural world.

Soft robotics has emerged as a promising alternative to rigid systems due to its ability to closely mimic biological functions (cite). Soft robots offer intrinsic safety, dexterity, and agility, while also enabling innovative solutions in sensing and movement. Advances in material science and fabrication techniques allow soft robots to integrate biological insights in ways that reduce computational demands—one of the key limitations of rigid systems (cite).

Despite this promise, most current soft robotic platforms remain focused on individual components—such as gripper design, vision-based sensing, or tactile feedback—rather than developing fully integrated, multifunctional systems (cite). This narrow focus has delayed their deployment in agriculture. Through the **ERISE initiative**, we are addressing this gap by proposing, **for the first time, a full-scale soft robotic**

solution from new soft robotic platforms to integrated plant sensing and phenotyping systems to fully deployable soft robots designed for both indoor and outdoor agricultural environments, all mounted on a mobile platform.

We propose to advance a new generation of soft, cognitively and morphologically adaptive agricultural robots—systems designed to learn, adapt, and operate safely across diverse biological and environmental conditions. Our **guiding hypothesis** is that soft-bodied robots, equipped with distributed, multifunctional, and multimodal sensing along with embodied intelligence, will outperform traditional rigid systems in complex, uncertain, and variable agricultural environments. These enhanced capabilities will enable novel functionalities in terrain navigation, in-situ diagnostics, and delicate crop manipulation—with agile and dexterous movement, reduced computational demands, and minimal risk of crop damage or safety concerns during operation.

Besides the aim of innovating high-end soft robotic systems, our broader vision is to make these technologies accessible to farmers—particularly small- and medium-scale growers. This goal aligns directly with the mission of ERISE and addresses a critical need in the Appalachian region, especially in WV, where over 75% of farms are small or medium-sized. Supporting these farms has the potential to transform the region's agrarian economy in the coming years.

To that end, we have chosen to focus on specialty crops, which are high-value and urgently require technological support. In **Phase 1** (Years 0–4), we will concentrate on tomatoes and bell peppers, which will serve as primary test crops for robot development. In **Phase 2** (Years 5–7), we will use the same robotic platforms to test on strawberries, thereby demonstrating the versatility and scalability of our system. These evaluations will be conducted in both indoor environments and challenging outdoor terrains, including mountainous regions.

The proposed robots will be designed to operate efficiently in both indoor and outdoor agricultural settings, with an emphasis on computational efficiency and real-time decision-making. Our **success metrics** include: 1) *Achieving robotic agility and dexterity through the fusion of tactile, vision, and proprioceptive sensing*; 2) *Integrating soft manipulators onto multi-terrain mobile platforms capable of detecting crop-specific diseases and phenotypes*; 3) *Enabling manipulation cycle times of under 15 seconds per fruit or vegetable, with a target success rate of over 80% for harvesting tasks through embodied intelligence*.

To achieve this vision, we have structured our research around four integrated focus areas:

1. **Embodied Intelligence and Distributed Sensing through Soft Robotics Design and Additive Manufacturing,**
2. **Learning in Coupled Biological–Mechanical Systems for Digital Twin-Driven Crop Modeling in Specialty Crops,**
3. **Plant Phenotyping, Disease Identification, and Crop Estimation for Soft Robotic Interaction**
4. **Decision-Making and General-Purpose Control in Soft Robotic Systems for Agile Harvesting.**

Focus Area 1: Embodied Intelligence and Distributed Sensing through Soft Robotics Design and Additive Manufacturing

While modern soft robotic systems offer clear advantages for agriculture, they remain limited by their modular, component-centric designs. Most platforms emphasize individual elements—like grippers or sensors—without achieving full system integration, leading to control challenges and insufficient adaptability in real farm environments. This research thrust seeks to advance embodied intelligence by integrating soft robotics, distributed sensing, and additive manufacturing. By embedding sensors directly into soft structures, robots can locally perceive and react to stimuli, reducing reliance on centralized computation. This distributed, compliant architecture will support terrain-capable movement and context-aware crop manipulation in unstructured agricultural settings.

What is urgently needed is a new class of soft robotic platforms that are **holistically designed and fully integrated**, capable of **embodied intelligence**, where the robot's physical structure, embedded sensors, and control strategies are co-designed to enable decentralized, real-time adaptation. To overcome control complexity, we propose to develop systems that couple **passive mechanical intelligence** (through compliant morphology and material properties) with **active, decentralized control** driven by distributed sensing and local processing. These robots will incorporate **multimodal sensing**—including tactile, force, proximity, and chemical—to safely and interactively engage with the variability and fragility of agricultural produce. To In this focus area, we synergistically focus on three key research objectives.

Objective 1.1. Bio-inspired Agile Soft Grippers with Integrated Multifunctional Sensing

Scientific Rationale: Grippers are one of the most critical components in agricultural robotic systems, serving as the primary interface for interacting with plants, fruits, and other produce during both pre- and post-harvest operations (cite). While grippers have been extensively studying ranging from rigid to soft design, current systems are often task-specific, lacking the agility and limited sensing required to match the nuanced capabilities of human grasping (cite). This severely limits their broader deployment in diverse farming environments. In this objective, we aim to develop a versatile bioinspired soft gripper that integrates multimodal sensing, including **tactile, chemical, and collocated vision sensors**, to support intelligent picking, in-field phenotyping, and crop distribution. By embedding sensing directly into the soft structure and enabling real-time stiffness modulation, our design will offer high agility and task generalization.

Research Plan and Methodology: We aim to develop an agile gripper using soft materials and additive manufacturing techniques, drawing inspiration from biological systems. Our design will integrate principles from biology, including distributed multifunctional sensing and mechanically intelligent structures, to enable adaptive, versatile grasping. To achieve this objective, we propose the following tasks:

Task 1.1.1: Design and develop bioinspired and origami-encoded soft grippers

We will develop a series of soft robotic grippers capable of touch-sensitive picking, leaf manipulation for plant phenotyping, pollination and disease sensing. These grippers will draw inspiration from biological systems and origami principles, particularly the Fin Ray structure (Fig. Xa–b). The Fin Ray effect is defined by passive adaptation, where the structure bends away from the point of contact rather than collapsing inward (cite). This enables a conformal grip around objects of varying shapes and sizes, reducing the need for complex control algorithms. Its cross-braced rib architecture supports compliant deformation, allowing gentle yet secure handling of soft or fragile produce—critical for obtaining accurate tactile feedback. The geometry of the Fin Ray structure also promotes self-alignment, increasing precision during operation in unstructured environments, and an example of built-in physical intelligence. Moreover, Fin Ray-inspired fingers can be agile and easily scaled and modularized, making them suitable for diverse agricultural targets. To enhance mechanical intelligence, we will further modify the rib structure using origami-based metamaterials. These embedded designs will allow the gripper to adapt its stiffness and grip force passively, responding to the mechanical properties of the object without requiring active force control.

The gripper design will feature cross brace ribs in the fin-ray configuration, which will allow the fingers to conform to object surfaces without causing damage. Fingers will be positioned 120° apart, providing three stable contact points to satisfy the minimum condition for a stable grasp [C]. Each finger will be actuated using either tendon-driven motor actuation or pneumatic control, while a central axis motor will allow for rotation of the gripper around its vertical axis for separation of the fruits or vegetables. The actuation unit will be located at the base of the finger plate to keep the system compact. To reduce control complexity and enhance modularity, we will also develop an independent actuation module that operates separately from the main manipulator (Fig. Xa). This modular design will facilitate plug-and-play integration with various robotic arms or mobile platforms. To achieve integrated and multifunctional soft robotic grippers, we will employ a combination of multi-material additive manufacturing and modular hardware assembly. The fabrication will be carried out in the PI Mishra's Robotics Lab, which is equipped with stereolithography (SLA) and extrusion-based 3D printing capabilities tailored for soft and rigid materials.

Task 1.1.2: Develop distributed multifunctional sensors

We will implement an integrated sensor design within the fin-ray gripper to enable multimodal environmental and mechanical feedback. Our goal is to embed at least four types of sensors directly into the gripper to support adaptive, autonomous interaction in unstructured agricultural environments. Most existing soft grippers incorporate either curvature or tactile sensors, and these are typically confined to specific regions. In contrast, we aim to achieve distributed, multimodal sensing across the entire gripper surface—an advancement that closely mimics the human hand and has not yet been realized in fin-ray architectures. One of the main challenges is integrating diverse sensing modalities into a compliant structure. To address this, we propose using light as the primary sensing medium. Light can be guided through soft optical materials, enabling signal transmission across the full length of the fingers, including regions that are otherwise inaccessible. This offers significant advantages in terms of sensor distribution, structural flexibility, and signal fidelity. Our approach will support proprioceptive, tactile, and chemical sensing, specifically tailored for agricultural tasks such as leaf and stem phenotyping and fruit harvesting.

Sensor Materials and Design: To develop the sensor, we will design both fiber-shaped and matrix-shaped structures using hydrogel materials. Fabrication will be performed via extrusion-based direct ink writing. These hydrogel sensors will be embedded within the gripper fingers to detect deformation based on changes in optical transmission. The hydrogel formulation will consist of gelatin dissolved in phosphate-

buffered saline (PBS), crosslinked with calcium chloride (CaCl_2), and photopolymerized using lithium phenyl (2,4,6-trimethylbenzoyl) phosphine (LAP) as the photoinitiator to enable UV-assisted curing and structural integrity during printing (cite). The printed hydrogel will act as a light-guiding medium, with infrared micro-LEDs serving as light sources and photodiodes positioned to detect signal variations in response to bending and deformation (Fig. Xa).

Tactile and Curvature Sensing: For tactile and curvature sensing, we will embed a finger-shaped sensor skin along both the dorsal and palmar sides of the gripper to ensure full surface coverage. For example, the palmar side of each finger will include an array of ten paired photodiodes and LEDs. Collectively, these pairs will provide information about bending (curvature), while individually they will detect localized deformation, enabling tactile force sensing.

Ethylene Gas Sensing: We will apply the same optical sensing principle to detect ethylene gas, a key indicator of fruit ripeness. In this case, hydrogel fibers will be arranged into circular bundles and functionalized with fluorescent molecular probes. These probes—such as transition metal-based fluorophores or ethylene-responsive supramolecular sensors—will be embedded in gelatin-based hydrogels and emit optical signals (fluorescence intensity or wavelength shift) upon ethylene binding (cite). Integrated micro-LEDs (UV-Vis range) and photodiodes will excite and detect these emissions, enabling real-time, non-contact sensing of ethylene concentrations for fruit phenotyping and plant stress monitoring.

Visual and Proximity Sensing: To enable high-level environmental perception and object interaction, we will integrate a commercial microcamera and a miniature distance sensor into the ray gripper system. The microcamera will support visual sensing for object recognition, pose estimation, and classification based on shape, color, and surface texture. The distance sensor will provide depth information for collision avoidance and spatial awareness. Together, these components will enhance grasp planning, target alignment, and post-grasp evaluation. Both sensors will be mounted on the wrist or base of the gripper to ensure a wide field of view without compromising the flexibility of the soft fingers.

Objective 1.2: A Multicellular Soft Arm with Self-growing, Touch & Proprioception

Scientific Rationale: In agricultural systems, the robotic arm plays a central role in interacting with complex plant canopies. While most existing platforms use rigid arms with 6 to 7 degrees of freedom and perform well in structured environments, they fall short in agricultural settings (cite). Tasks like repositioning branches, uncovering occluded fruit, and pruning occur in cluttered, deformable, and partially observable environments, where rigid arms often lack the compliance and tactile sensitivity needed. This mismatch can result in excessive force application, crop damage, or task failure. To address these limitations, we propose the development of a fully soft continuum robotic arm designed for adaptive and safe interaction with flexible plant structures. By leveraging the coupled deformation dynamics between the arm and the canopy, the system will enable intelligent, natural responses. Key features of the proposed arm include: (1) Mechanically intelligent obstacle avoidance for passive navigation around branches and leaves without active sensing; (2) Integrated proprioceptive and tactile sensing for real-time, sensor-informed control; (3) High-load capacity soft structures to support tools and end-effectors for operations such as picking, grasping, phenotyping; and (4) Agile and dexterous motion to access occluded or hard-to-reach targets. Targets.

Research Plan and Methods: We aim to develop a soft growing robotic arm inspired by biological structures such as vines, roots, and tentacles (cite). By exploring emergent behavior through local rules and decentralized control, this work addresses fundamental questions in soft manipulation: how can modularity and morphological computation enable generalization across objects and tasks without centralized planning? It will produce new architectures for soft robotic control, contribute to scalable manipulation strategies, and demonstrate principles of physical intelligence applicable to a broad class of agricultural and non-agricultural systems and applications. To achieve this goal, we will focus on designing the continuum structure, developing embedded sensing, implementing task-specific motion patterns, and integrating end-effectors for direct canopy interaction.

Task 1.2.1: Multicellular self-growing soft continuum arm with integrated shape and tentacle sensing.

We will focus on developing a soft robotic arm inspired by growing and multicellular mechanisms, designed to adaptively interact with plants in complex agricultural environments. The proposed system consists of modular soft manipulators capable of performing tasks such as screening, holding, and picking plant stems or peduncles. The arm's architecture is based on a multicellular continuum design, where each cell provides localized movement for high dexterity and adaptability. multicellular soft arm that draws from the Loopy design to achieve real-time reconfiguration and general-purpose adaptability through distributed

intelligence (Fig. x). The proposed arm will be embedded with distributed force and contact sensors, enabling it to estimate the shape, stiffness, and response of entangled crops during manipulation. Using reservoir computing and system identification, the robot will learn to predict how different motions influence both the arm and plant, allowing for closed-loop control of compliant, safe interactions. The main inspiration here is treating both the arm and the plant branches as soft robots. We can then use similar modeling, prediction, simulation, and control tools for both. Unlike rigid systems that depend on global state estimation and exact models, this approach leverages local feedback and learning to infer the hidden state of the interaction in real time (see Fig. x).

As shown in Fig. Xa, the arm consists of five modular units (M1–M5), each containing three chambered soft actuators capable of bending and elongation for precise shape and position control. Inspired by the growth behavior of tendrils and vines, this design enables dynamic extension and shape adaptation. To manage the complexity of controlling such a flexible structure, we adopt a growing-inspired, master–follower control strategy. As illustrated in Fig. 2c,

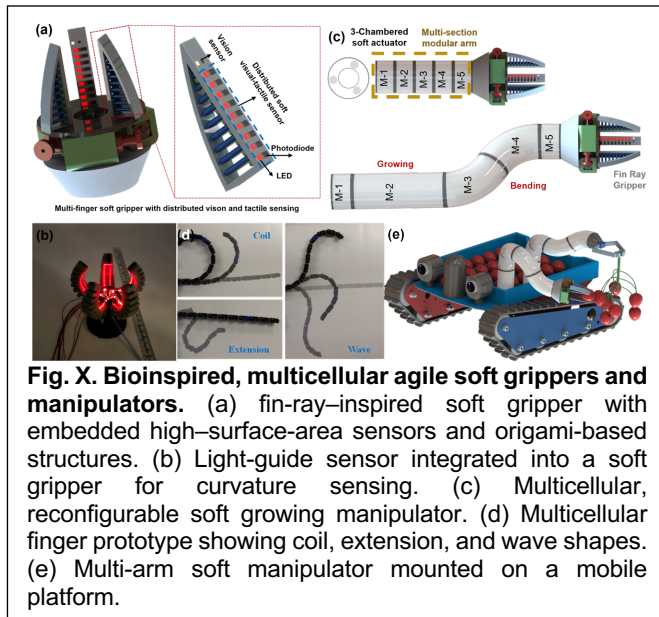


Fig. X. Bioinspired, multicellular agile soft grippers and manipulators. (a) fin-ray-inspired soft gripper with embedded high-surface-area sensors and origami-based structures. (b) Light-guide sensor integrated into a soft gripper for curvature sensing. (c) Multicellular, reconfigurable soft growing manipulator. (d) Multicellular finger prototype showing coil, extension, and wave shapes. (e) Multi-arm soft manipulator mounted on a mobile platform.

modules activate sequentially (M1 to M5), producing telescopic motion that enhances reach, stability, and passive compliance—ideal for navigating cluttered environments. To support autonomous and accurate motion, we will integrate soft curvature and bending sensors for shape estimation, along with tactile sensors to detect interaction forces with plant structures. These sensing elements will be embedded within the soft body to provide real-time environmental feedback, improving safety and effectiveness. The arm modules will be fabricated using the same techniques described in **Task 1.1.1**. Each unit will be made from a silicone-based elastomer capable of over 300% strain and 3.2 MPa tensile strength, offering durability and flexibility for high-deformation tasks. Modules will be connected via elastic tubing and routed tendons, allowing for hybrid actuation: pneumatic chambers enable

extension through controlled inflation, while tendon-driven actuation—using geared DC motors and fishing-line tendons—modulates bending and stiffness. For sensing, we will extend the approach described in **Task 1.1.2** by embedding optical light-guiding hydrogel fibers with integrated micro-LEDs and photodiodes. These sensors will enable real-time shape sensing and tactile feedback, enhancing interaction with stems and plant canopies.

Objective 1.3.: Adaptive Soft Whegs with Multicellular Intelligence for Terrain-Aware Mobility

Scientific Rationale: Agricultural terrain is unpredictable ranging from steep, uneven slopes, muddy soils, and cluttered environments. While wheel-leg hybrids (whegs) have demonstrated utility in unstructured environments, their current forms are largely passive, lacking the sensing and intelligence required to adapt to environmental changes in real time (cite). Additionally, these types of robots will play a critical role in WV, where much of the outdoor agriculture takes place in unstructured and uneven terrain rather than flat farmland. In such environment's, purely wheeled systems are prone to failure, underscoring the need for a new class of mobile robotic platforms. This research proposes a next-generation wheg system that integrates soft robotic design with a multicellular architecture—building on the Loopy robot—to enable terrain-aware, adaptive mobility. It addresses a foundational question in soft robotics: how can multicellular control architectures allow robots to reason about and respond to environmental variability through embodied adaptation?

Research Plan and Methods: We plan to develop a wheel and leg fused robotic platform by leveraging principles of physical intelligence and multicellular-inspired organization to achieve robust, terrain-adaptive locomotion. This hybrid platform will combine the speed and efficiency of wheeled systems with the terrain versatility and stability of legged mechanisms, enabling operation across diverse agricultural environments. The soft growing arm developed in previous tasks will be mounted on this mobile platform (Fig. X), allowing for field deployment and autonomous navigation in real-world agricultural settings. The integration of

adaptive locomotion with morphologically intelligent manipulation will result in a highly versatile system capable of performing tasks such as in situ harvesting, pollination, phenotyping, and disease detection.

Task 1.3.1. Development of Multi-Terrain Whegs Robots for Outdoor and Indoor Mobility

The current wheg design for WVU's University Rover Challenge (URC) has three components: rigid aluminum spoke, flexible carbon fiber/kevlar composite footer, and soft TPU wheg "socks" that cover the footer and provide shock absorption with its material property and geometric design. The proposed wheg design consists of modular soft toes, each functioning as an independent sensory-actuation unit with embedded tactile sensors, pressure sensors, and local computation. These cells coordinate through distributed control algorithms inspired by morphogen-based signaling, adjusting stiffness, gait, and contact behavior in response to sensed terrain properties. This architecture enables the system to adapt dynamically to different surfaces, reduce slippage, and maintain stability without reliance on central control or precise terrain modeling.

The research will investigate how distributed physical intelligence can emerge from local sensing and actuation, and how soft morphological structures influence locomotion performance in high-uncertainty conditions. It will also contribute to new models of terrain classification, energy-efficient movement, and adaptive control for compliant mobile systems.

Preliminary results: PI Mishra has developed an underactuated five-fingered soft gripper (Fig. XX) using pneumatic actuation and integrated tactile and vision sensing. The gripper can handle delicate fruits such as strawberries and includes optoelectronic light-guide sensors embedded in each finger for real-time tactile feedback. Preliminary results demonstrate that this sensor fusion enables accurate fruit detection, ripeness classification, and distinction between soft produce (e.g., strawberries) and firmer fruits (e.g., avocados) (Fig. X). In addition, PI Mishra has designed a reconfigurable soft gripper actuated by a tendon-driven motor system, capable of adapting its morphology to various object shapes and stiffnesses. This gripper has safely handled diverse items, including ripe pears, tomatoes, eggs, and glass bulbs (Fig. X). As part of his contributions to soft robotic manipulation, PI Mishra developed a soft manipulator called SIMBA (Soft Intelligent Modular Bioinspired Arm), published in *Frontiers in Robotics and AI*. SIMBA demonstrated high dexterity and adaptive shape control using tendon-driven actuation and was successfully tested in unstructured environments for tasks such as tomato harvesting and door opening (cite, Fig. XX). Additionally, PI Mishra's prior work on hydrogel-based soft actuators, published in *Science Robotics*, will serve as the foundation for developing hydrogel-based light-guide sensors to be embedded in future grippers and manipulators. PI Gu has developed multicellular robotic platform loopy (cite) that consists of 36 connected servos and has led two WVU URC teams to 2nd place overall finished with wheg-robot designs.

Expected Outcomes: This project will deliver a modular, bio-inspired soft robotic system that integrates adaptive manipulation, multifunctional sensing, and terrain-aware mobility to meet the needs of modern agriculture. Key outcomes include: (1) a fin-ray soft gripper with distributed multimodal sensing and sub-second grasping speed; (2) a multicellular soft continuum arm with dexterous manipulation, passive obstacle avoidance, and integrated tactile and proprioceptive sensing, achieving task execution within 10 seconds; and (3) integration of multiple soft arms onto a multi-terrain mobile platform with full grasping and sensing capabilities. These efforts will converge into a unified platform for general-purpose agricultural robotics. The resulting system will embody the principles of softness, decentralized intelligence, and biological adaptability, enabling safe and autonomous operation in small, mixed-use farms. It will also serve as a powerful testbed for advancing whole-body coordination, low-energy control strategies, and adaptive task generalization in unstructured environments.

Focus Area 2: Learning in Coupled Biological–Mechanical Systems for Digital Twin-Driven Crop Modeling in Specialty Crops

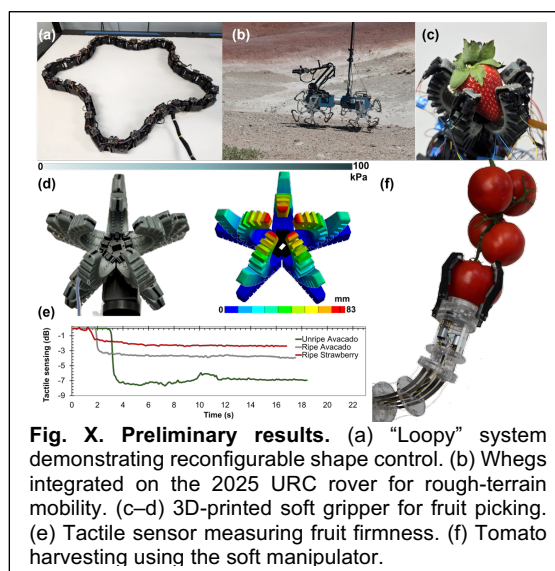


Fig. X. Preliminary results. (a) “Loopy” system demonstrating reconfigurable shape control. (b) Whegs integrated on the 2025 URC rover for rough-terrain mobility. (c–d) 3D-printed soft gripper for fruit picking. (e) Tactile sensor measuring fruit firmness. (f) Tomato harvesting using the soft manipulator.

The current field agricultural robotics face several critical challenges, particularly due to the dynamic and variable nature of plants, which differ significantly across indoor and outdoor environments. In crops with soft canopies, fruit location and posture constantly shift due to gravity, wind, and external disturbances—including interactions from humans or robots. These subtle but continuous changes make stable and reliable grasping difficult, especially as fruit stems bend, twist, or detach unpredictably. Additionally, there is significant variability in plant morphology and physiology—including leaf and fruit color, size and shape, and mechanical properties such as stem or fruit stiffness, all of which change over time and with environmental conditions. These dynamics create a moving target for robotic systems and are a major reason why many modern agricultural robots lack efficient manipulation capabilities, often resulting in slow performance and limited deployment in real-world settings.

To address this, our focus is to better understand the physical and mechanical behavior of plants and their interaction with the environment. Our goal is to develop one of the most versatile, physics-informed digital models of three specialty crops—tomatoes, bell peppers (phase 2), and strawberries (phase 2)—by characterizing their mechanical, optical, and geometric features. These digital twin models will serve as high-fidelity simulators to guide the development of agile, human-like soft robotic manipulators capable of operating in complex crop environments. In the first four years, we will focus primarily on bell peppers and tomatoes, two high-value crops grown extensively in WV, with over 600 high tunnels supporting their production across the state. We will pair selective breeding of commercial varieties with detailed biophysical analysis to build a robust, adaptable digital twin framework that bridges plant biology and robotic control—ultimately enabling efficient detection, interaction, and harvesting under real-world conditions. To realize the goals outlined in **Focus Area 2**, we will pursue **three key research objectives**, each addressing critical aspects of plant dynamics, sensing, and robotic control.

Objective 2.1: Develop Conventional Fast Breeding Methods for Tomatoes, Bell Peppers, and Strawberries for Robotic Indoor and Outdoor Testing

Scientific Rationale: This objective focuses on establishing accelerated breeding protocols for key specialty crops—tomatoes, bell peppers, and strawberries—tailored for integration with robotic testing platforms. Selected cultivars will be optimized for controlled indoor environments and semi-structured outdoor settings to enable reliable robotic manipulation, phenotyping, and harvesting trials. This work will create a standardized pipeline for generating plant lines suited for mechanical interaction, including traits such as uniform geometry, stem structure, and fruit attachment. Use conventional plant breeding methods to select for traits such as reduced leaf density, upright fruit growth, uniform cluster formation, and pruning responsiveness.

Research Plan and Methods:

Task 2.1.1. Tomatoes breeding

SOFT-AG will use an eight-month greenhouse tomato crop as its primary “living testbed” for physics-informed digital twins and soft-robotic harvesters. Two indeterminate cultivars are recommended that will provide contrasting architectures for algorithm generalization: Dormaplum, a snack-plum line with short pedicels and firm pericarp proven compatible with automated pickers (4), and BHN 589, a 220-g salad tomato valued for its uniform fruit and moderate canopy that simplifies machine vision (5). Seeds will be sown in sterile peat–perlite (pH 6.5–6.8) at 24 °C/20 °C and 90 % RH; WVU Extension guidelines recommend supplying early supplemental light to prevent etiolation (6). After approximately 28 days, transplants will be moved to 10-L aerated coco-coir slabs (1.2 m × 0.4 m spacing) under 24/18 °C, 70 % RH, and drip fertigation at EC 2.5 dS m⁻¹. This regime speeds vegetative growth and flowering without weakening stems (7). Plants will be trained to single-leader high-wire systems lowered and leaned weekly to a height of 3.2 m. Anthesis appears by Day 35; mature-green fruit develops approximately 55 days post-flower and reach breaker stage approximately Day 88. Each pick will log plant, location, detachment force, and stem deflection; pedicels will be clipped to <8 mm for uniform gripping (8). Depending on available space, four randomized blocks will be used (n = 64) to ensure spatial replication. The sample space may be adjusted to accommodate space constraints. By trial’s end, the study will deliver high-resolution morphological and biomechanical data, validated low-cost sensors, cultivar-specific harvestability metrics, and progress in integrating greenhouse crops and autonomous manipulation.

Task 2.1.2. Bell pepper breeding

SOFT-AG will also integrate sweet bell peppers (*Capsicum annuum*) into its digital-twin and soft-robotics platform to diversify crop morphology and manipulation challenges. Two high-wire greenhouse cultivars are recommended to be grown: ‘Early Sunsation’, a yellow 180-200 g blocky bell with upright, easily imaged pedicels, and ‘Autry’, a jumbo red type whose thick walls and high detachment force stress gripper-

compliance algorithms (9). Seeds will be started eight weeks before transplant, the longest indoor window among common vegetables (10), in sterile peat-perlite (pH 6.5–6.8) at 27 °C/23 °C and 95 % RH. A 16-h photoperiod supplying 18 mol m² d⁻¹ and a 2.5 dS m⁻¹ nutrient solution produces stout, uniform transplants (11). At approximately 56 days after sowing, it is recommended that plants be moved to 10-L aerated coco-coir slabs spaced 1.2 m between gutters and 0.45 m in-row, slightly wider than tomato placement to accommodate broader canopies. Greenhouse set-points will mirror concurrent tomato trials (24/18 °C, 70% RH, 1,000 ppm CO₂). However, LEDs shift from a 3:1 to a 5:1 red: blue ratio at fruit set, boosting marketable yield by approximately 2% in bell peppers (12). Each plant will be trained to a bifurcated Spanish-V trellis topped at 2.8 m; weekly lower-and-lean operations and selective leaf removal create a planar fruiting wall that simplifies robotic path planning. Anthesis occurs approximately 28 days post-transplant; the mature-green stage follows in 30–35 days, and full color is achieved in 35–45 days, allowing for continuous harvest from Day 63 to Day 210. During each 2 to 3 day pick cycle, fruit ID, 3-D coordinates, detachment force, and cycle time will be recorded with the aim to achieve a <2% damage rate, as reported for commercial pepper robots (13) (14). By trial's end (~7 months), the crop should deliver approximately 8 kg m² of graded fruit, > 1,200 annotated pick event opportunities, and high-resolution biomechanical data, expanding E-RISE's tri-crop dataset and defining robot-readiness traits for next-generation pepper breeding.

Task 2.1.3 Environmental biophysics of especialviony crops

To ground digital twins and robotic harvesters in rigorously measured reality, each 60 m² bay (space contingent) used for the current proposed research in the WVU Evansdale Greenhouse will be equipped with a lean, environmental-biophysics sensor suite. Aspirated temperature–relative-humidity probes (± 0.1 °C, ± 1 % RH) mounted at canopy top and mid-canopy, quantum sensors for photosynthetically active radiation, and a four-component net radiometer will capture the energy terms that drive plant water loss and carbon gain, core variables highlighted in environmental-biophysics theory (15). Two capacitance probes per root-zone slab will continuously log volumetric water content and matric potential. A precision weighing lysimeter, located beneath one representative slab per crop, will supply ground-truth evapotranspiration (ET_{actual}). Paired infrared thermometers will be used to record sunlit and shaded leaf temperatures, allowing for the computation of leaf-to-air vapor-pressure deficit, a direct proxy for stomatal conductance, every minute (16). Reference ET will be solved with the FAO-56 Penman–Monteith equation and updated daily by calibrating crop coefficients to lysimeter output. This greenhouse parameterization has achieved <0.3 mm d⁻¹ RMSE in tomato studies {Citation} and improved water productivity in South-Korean bell-pepper trials (Yang[JH4] et al. 2024). When modeled and measured ET diverge by more than 0.5 mm d⁻¹, drip-irrigation pulses will be adjusted to maintain slab matric potentials above –15 kPa (for tomatoes) and –20 kPa (for peppers), an approach expected to reduce water use by approximately 12 % without yield loss (Yang[JH5] et al. 2024). Crop condition will be assessed with two complementary non-contact tools. Narrow-band fluorometers fixed to sentinel leaves track the maximal fluorescence (F_m) and the fluorescence-decline ratio (R_{fd}), metrics that closely correlate with tomato ripening, with R² value of greater than 0.96 (Abdelhamid[JH6] et al. 2021). Every three days, an overhead rail carrying twin 24-MP RGB cameras photographs the crop; Structure-from-Motion reconstruction, aided by a kaolin particle-film anti-glare spray, creates centimeter-scale meshes whose fruit-volume and colour-index errors are <5 % (Masoudi[JH7] et al. 2024). Mesh-derived canopy height, leaf-area index, and fruit hue angles will assist the digital twin, triggering harvest orders once peppers or tomatoes reach maturity thresholds established by a three-band VNIR classifier that achieves over 90 % accuracy (Muñoz[JH8] -Postigo et al. 2023). The study design will provide minute-resolution ET with less than 0.3 mm d⁻¹ error, over 100 time-stamped 3D models per plant, and at least 85% accurate identification of breaker-stage tomatoes and full-color peppers, data dense enough to enhance reinforcement-learning harvest policies, yet produced by hardware that is affordable, scalable, and grounded in environmental-biophysics fundamentals.

Objective 2.2: Develop Biophysical Interfaces with Pepper and Tomato Plants Through Multi-Sensor Fusion

Scientific Rationale: Plants are highly dynamic and deformable living systems, and one critical yet underexplored component in agricultural robotics is the force interaction between plants and robots, especially in soft–soft body contact scenarios. Plant stems, leaves, and fruits are not static; they deform under their own weight, respond to external mechanical stimuli, and reorient dynamically due to factors like wind, growth, or human and robotic interaction. However, our current understanding of the biomechanical responses of plants during robotic manipulation is limited. To enable more precise and efficient manipulation, it is essential to study the mechanical coupling between soft robotic actuators and the deformable plant structures they interact with. Understanding how two compliant bodies interact and

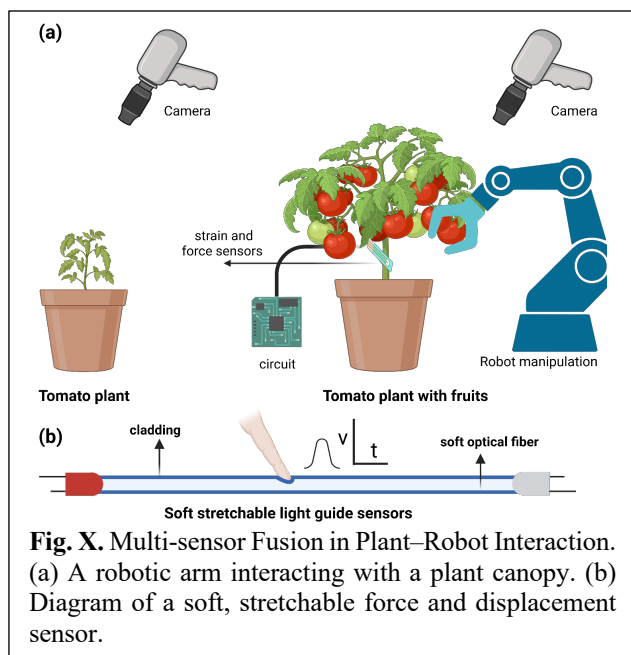


Fig. X. Multi-sensor Fusion in Plant–Robot Interaction. (a) A robotic arm interacting with a plant canopy. (b) Diagram of a soft, stretchable force and displacement sensor.

respond to contact forces is key to reducing grasping cycle times and preventing damage to delicate tissues.

Research Plan and Methods: In this objective, we aim to develop biophysical interfaces that allow soft robotic systems to interact intelligently and safely with pepper and tomato plants. These interfaces will capture force-displacement relationships, structural adaptation, and dynamic responses during robotic contact, forming the basis for accurate physics-based models that inform real-time adaptive control strategies. We will achieve this through the following tasks.

Task 2.2.1. Sensor Design and Integration into Plant Canopies

As we plan to study the nature for plant behavior We will develop a stretchable optical fiber sensor system designed for seamless integration into plant branches. Specifically, we aim to fully sensorize individual plants by embedding hundreds of low-cost, flexible sensors throughout the canopy and stem structures. These sensors

will be fabricated using Stretch Magic™ urethane-based stretchable optical fibers, combined with infrared LEDs and photodiodes to enable precise displacement and force readouts (Fig. x). Our goal is to measure plant deformation under two key conditions: (1) during robotic manipulation and external disturbances such as wind or mechanical contact, and (2) during the growth progression from early development to fruit maturity. We hypothesize that branch reconfiguration—caused by either fruit loading at maturity or external interaction—alters fruit position dynamically, requiring the robot to repeatedly re-estimate its targets. By embedding real-time sensing into the plant, we can anticipate these positional changes and autonomously update robotic control algorithms, improving precision and efficiency. In addition to sensor deployment, we will develop simple biomechanical models to predict branch deformation behavior during pruning, picking, and packaging processes. These models will help refine control strategies and provide a foundation for future digital twin integration of plant–robot interactions.

To study the temporal changes in plant growth, fruit development, and stem geometry, we will employ a photogrammetry-based 3D reconstruction pipeline using Agisoft Metashape. A 360-degree camera will be mounted on a fixed tripod while the potted plant is placed on a motorized rotating platform. The rotation stage will allow for consistent, controlled angular increments, capturing a full set of high-resolution images every hour. This process will be automated and repeated hourly to generate time-series data capturing plant growth dynamics. The reconstructed 3D models will allow for quantitative analysis of 1) Fruit size and curvature changes; 2) Stem bending and reorientation; 3) Canopy deformation due to environmental or robotic interaction. Post-processing and model comparison will be performed using CloudCompare and custom Python scripts to track geometric changes across time points. This setup will also support the development of digital twin models by integrating morphological data with robotic control frameworks.

Objective 2.3: Develop Digital Twins with Biophysics-Integrated Parameters for Precise Robotic Manipulation and Phenotyping

Scientific Rationale: This objective will establish digital twins of crop–robot interaction systems that incorporate biophysical properties of plants, sensor feedback, and environmental variables. These virtual models will simulate plant behavior and robotic actuation dynamics to support precise manipulation, adaptive control, and automated phenotyping. The digital twins will serve as a tool for real-time strategy optimization, predictive decision-making, and robot training across various crop types and growth stages. Create digital phenotype libraries using sensor-collected data (Focus 3) to accelerate selection cycles.

Research Plan and Methods:

Task 2.3.1. UAV–UGV Multimodal Data Fusion & Crop Digital Twins

To develop crop-level digital twins, this task will explore the fusion of multi-modal aerial and ground robot data to enhance model reconstruction. We will build upon pipelines such as AgriColMap 3D Mapping pipeline {ref}, which have been shown effective a leveraging photogrammetric-base reconstruction using

both unmanned aerial vehicle (UAV) and unmanned ground vehicle (UGV) data to include multimodal sensors. UGV's additional payload and power capacity allows additional sensors, such as 3D LiDAR, which can be used to enhance the framework and improve reconstruction quality. A LiDAR-inertial SLAM Georeferenced with GPS data will be used on the UGV to estimate a sparse 3D map of the environment. Next, fine image data from both UAV and UGV will be used to reconstruct 3D maps of the field. LiDAR and image-based UGV maps will be registered via Deep Learning based matching using a pipeline like what has been shown effective for terrestrial applications. The combined point-cloud of the UGV will be registered with that of the UAV with ArgriColMap. Our research will explore 1) constraining the registration problem through estimated cooperative localization between the UAV and UGV, and 2) enhancing mapping quality by considering the cooperative motion planning of the UAV and UGV. This will lay the groundwork for actionable situation awareness in the crop fields.

Preliminary results: PI Mishra has developed an inexpensive, scalable sensor system costing less than \$10 per unit, which is currently under review (Fig. Xd). We plan to deploy this sensor design to sensorize tomato and bell pepper plants, enabling real-time monitoring of plant deformation, fruit loading, and branch dynamics. These low-cost sensors will provide force and displacement data critical for adaptive robotic control during harvesting. In addition, PI Mishra has previously developed a high-resolution 3D reconstruction method for capturing detailed shape and color information of plant roots and vegetables, as reported in [ref]. Building on that work, we will apply an improved version of this model to above-ground plant structures, enabling precise localization and planning for fruit grasping tasks. The enhanced design will incorporate photogrammetry and multisensor fusion to support robotic perception and decision-making in complex agricultural environments.

PI Gross has extensive prior work on mapping and cooperative localization that will be leveraged in this work. in Task 2.3.1. His students have considered UAV trajectory planning for quality mapping of limestone pillars {ref}, cooperative localization and belief state planning of UAV and UGV in subterranean environments {ref, ref, ref} and point cloud registration from UAV maps {ref}

Focus Area 3: Plant Phenotyping, Disease Identification, and Crop Estimation through Multimodal Sensing and Artificial Intelligence for Soft Robotic Interaction

This research thrust will develop robust perception and estimation frameworks to infer crop state, mechanical properties, and spatial structure under occlusion, clutter, and variable lighting. By creating methods that fuse RGB-D, LiDAR, hyperspectral, thermal, and tactile data into actionable 3D crop maps and digital phenotypes, we will address the modeling challenges posed by soft robotic sensors and actuators, which often introduce nonlinearity, drift, and deformation-dependent noise, requiring new approaches in active sensing, uncertainty modeling, and real-time state estimation.

Objective 3.1: Robust Multi-Modal Fruit Detection and Localization based on 3D Reconstruction

Scientific Rationale: Conventional machine vision for robotic harvesting often struggles even with foliage camouflaging fruit, as well as shadows and bright sunlight {ref}. For example, immature green fruits can share color with leaves, as this defeats RGB-only detection {ref}. Sensor fusion with multiple modes provides complementary information {ref}. This information fusion is designed to address these issues. For revealing hidden targets, thermal imaging highlights temperature contrasts, whereas an RGB camera provides high-resolution color cues, as fruits cool or warm at different rates than leaves. Even when colors appear similar, hyperspectral imaging adds narrow-band spectral signatures to distinguish fruit from surrounding vegetation {ref}. Because the system merges these modalities, a richer scene representation is obtained {ref}. Real orchards have strong detection with better fruit visibility. Past studies show promise: Gan et al. {ref} detected green citrus at ~87% precision by using “active” thermal imaging (spraying cool mist to induce fruit-leaf temperature differences), which made fruits stand out thermally. Likewise, fruit localization accuracy in conditions of occlusion has been improved by combining stereo/RGB-D vision with spectral data {ref}. Sensor fusion reliably detects fruit outdoors within unstructured settings, as these findings show.

Research Plan & Methods: A multi-sensor device including synchronized RGB and thermal cameras plus a small hyperspectral imager will be co-registered spatially and temporally. The fused perception pipeline will use a deep learning model tailored for multimodal input. We will use lightweight architecture with feature fusion techniques. Embedded GPUs will allow inference in real time using the techniques. The innovation lies in expanding the perception bandwidth: fusing heterogeneous data to detect fruits that single-modality detectors would miss (e.g. fruit behind leaves or color-blending fruit). The other approach is to reconstruct the 3D scene of the plants using Gaussian splatting methods {ref}. An example of the Gaussian splatting-based flower pose estimation method in Fig. ZZ.

The 3D scene of the plant will be generated by the soft robot developed in **Focus Area 3** to plan the movement trajectory for fruit picking. To carry out this research, the following tasks will be performed.

Task 3.1.1: Field Data Acquisition, Annotation and Deep Learning-Based Feature Fusion

After hardware integration of the sensors on the robot, multimodal data that is wide-ranging will be collected from test plots of tomatoes and bell peppers in outdoor fields and high tunnels. Camera calibration and extrinsic calibration between the multi-modal sensors will be conducted before collection. Collections will be conducted in various lighting conditions, growth stages, and occlusion scenarios. Each set of synchronized RGB, thermal, as well as hyperspectral images will be labeled either manually or semi-automatically for fruit presence. Maturity level in addition to occlusion status will also be labeled. These annotated datasets will serve both in model training and benchmarking because they offer real-world variability important for generalization.

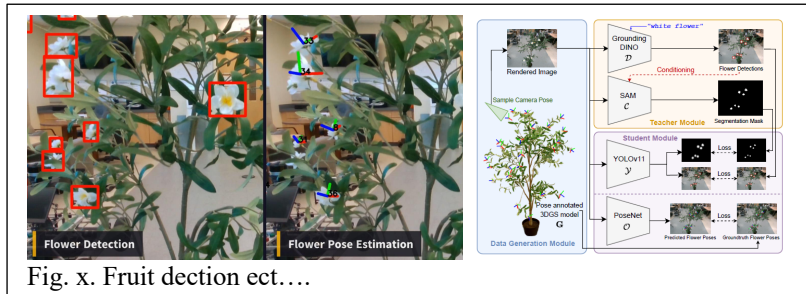


Fig. x. Fruit detection ect....

The team will then design a deep learning architecture for fruit detection accepting RGB, thermal, and hyperspectral inputs. We will experiment about different fusion strategies. We will combine outputs of modality-specific detectors late, merge intermediate feature maps mid-level, and concatenate raw input channels early. The architecture will be

tailored to deployment on embedded platforms such as the NVIDIA Jetson series, plus it uses lightweight backbones like EfficientNet {ref} or YOLOv8 nano {ref}. This real-time inference will preserve accuracy across a range of field conditions.

Task 3.1.2: Model Training, Validation, Optimization, and Deployment

The fused detection model will be trained upon the annotated dataset because we are using standard metrics such as precision, recall, also F1-score. Robustness is going to be improved using augmentation techniques. We will use occlusion simulation, motion blur, and illumination variation. To assess generalization, tests within unseen environments will need validation. The model will then be optimized for real-time use because it will balance accuracy with computational efficiency. As needed, the team will explore reducing latency and memory footprint by using techniques like quantization, pruning, or tensor decomposition.

The final step involves the deployment of the trained model within operational farm environments located on the robotic platform. Real-time performance will be assessed through metrics like frame rate and detection accuracy under occlusion plus robustness to environmental changes like wind-blown foliage or glare. Comparisons against single-sensor baselines will show multimodal fusion's advantages. Baselines might be RGB-only and thermal-only.

Objective 3.2: Multi-Sensor Ripeness Estimation for Harvest Timing Decisions

Scientific Rationale: Multi-sensor ripeness estimation will enable soft robotic harvesters to autonomously distinguish ripe produce from unripe fruit or overripe fruit. Hyperspectral imaging analyzes narrow-band reflectance patterns invisible to standard RGB imagery for capturing subtle biochemical changes like chlorophyll breakdown or sugar accumulation. In situ hyperspectral systems recently demonstrated near perfect accuracy classifying ripeness stages of citrus and strawberries under field conditions {ref}. Tactile sensing is found within soft grippers for measurement of flesh firmness as it gently grasps {ref}. This assessment is how humans assess ripeness via touch. The system we use combines optical clues from afar like near-infrared spectral indices with mechanical feedback near to hand like deformation dynamics. This gives a complete ripeness review by combining these methods instantly. Current methods often rely on single modality sensing or offline inspection, but as we integrate, the robot will autonomously decide about whether to harvest each fruit because it measures confidence uniformly

Research Plan & Methods: The gripper will use a small hyperspectral camera alongside a tactile sensor for this objective. A two-step model will integrate in-grasp sensing with pre-grasp remote sensing through a fusion algorithm that will combine optical data and tactile data into a final decision. The following tasks will be performed to accomplish this:

Task 3.2.1: Optical & Hyperspectral Sensing Module Integration

This task will embed a compact multispectral imaging unit within the robot's gripper that combines RGB and key narrow-band sensors like ~750 nm or ~1,400 nm. Deep learning architectures outperform vision-only baselines since they draw on hyperspectral ripeness classification of strawberries and avocados {ref}. The module eases assessment of biochemical ripening indicators like chlorophyll decay or sugar accumulation, and that assessment will not be destructive. The miniaturized sensor system will be optimized for rapid capture and low power, so it can be used in real-world fruit-harvesting robots deployed at the edge.

Task 3.2.2: Vision-Based Tactile Firmness Sensor Development

Next, a tactile sensor like a camera-equipped soft-finger will be integrated such as the (GelSight Fin-Ray){ref} or deformation-based pressure pads into the soft gripper so as to directly measure fruit firmness. For example, Lin et al.'s work achieved ~0.79 R^2 firmness prediction as well as ~84 % maturity grading accuracy within soft robotic testing, plus it then inspired our own work {ref}. This sensor will capture therefore force-deformation profiles during grasp. Tactile analysis in real-time will help optical assessment because it finds internal maturity that changes like softening unseen visually.

Task 3.2.3: Multimodal Fusion Model Development

Paired optical and tactile multispectral data ground truth data will then be collected while ranging across ripeness levels in orchards like tomatoes or strawberries to support model development {ref}. This will assist in developing a classification model for estimation of ripeness with an associated confidence score, that combines both optical and tactile inputs, such as either a Bayesian network or a multi-input deep neural network. Capsule or probabilistic models used in tactile perception inspire this methodology which will weight sensory confidence along dynamically, prioritizing optical metrics early and tactile data during grasp. This fusion improves robustness over single-modality approaches.

Task 3.2.4: Edge Inference Optimization, Control Integration and Testing

Finally, we will optimize the fusion model toward real-time operation on a compact embedded system like NVIDIA Jetson or a microcontroller with neural accelerator. Latency will decrease as performance persists using methods such as pruning or weight quantization. The control system is going to then use that ripeness confidence score to autonomously execute harvest actions such as when grasping only fully ripe items or else releasing others. This enables the real-time decision-making in picking missions. We will then deploy the complete system in WV farm environments with the goal is for decision accuracy above 90 % and minimal bruising.

Objective 3.3: Vision–Language Interpretation of Multimodal Data for Task Planning and Decision Support in Agricultural Robotics

Scientific Rationale: Agricultural environments present unique challenges for robotic perception and action: fruit is often partially occluded by branches, lighting conditions vary drastically, and plant structures are irregular and dynamic. These conditions limit the generalizability and performance of traditional vision models, which require extensive labeled datasets and retraining for new contexts. Recent advances in foundation models, including segmentation models such as Segment Anything Model (SAM) {ref}, vision–language models (VLMs), e.g., CLIP {ref}, and large language models (LLMs), e.g., GPT-4 {ref} and Claude 3, offer a transformative pathway toward context-aware, zero-shot, or few-shot, and language-guided robotic intelligence.

SAM enables zero-shot segmentation based on visual prompts, even in novel settings, making it well-suited for segmenting fruits and branches in visually complex agricultural scenes. Meanwhile, VLMs and LLMs can reason about high-level task goals, translate fused sensor data into actionable summaries, and generate multi-step plans in natural language. By integrating these capabilities, this research thrust aims to create intelligent systems that bridge low-level perception and high-level reasoning for adaptive, semantically informed harvesting and decision-making.

Research Plan and Methods:

Task 3.3.1: Zero-Shot Fruit Localization via Prompt-Based Segmentation and Visual Detection

We will leverage SAM as a foundational segmentation model to support zero-shot detection of fruits in wild and cluttered environments. Unlike traditional object detectors, SAM enables segmentation using spatial prompts (e.g., bounding boxes or points) and generalizes across object types. This capability is particularly valuable when occlusions from leaves, inconsistent lighting, or fruit variability make conventional model training impractical. To improve accuracy and runtime efficiency, we will integrate SAM outputs with lightweight, domain-adapted You Only Look Once (YOLO) models {ref-original YOLO} trained on datasets collected from tomatoes and bell peppers. SAM will serve as a promptable segmentation guide in complex

scenes, while YOLO will provide efficient real-time detection in well-characterized scenarios. This hybrid system will be benchmarked for robustness, precision, and speed under different environmental conditions.

Task 3.3.2: Vision–Language Task Planning for Soft Robot Manipulation

We will implement a task planning pipeline driven by LLMs, e.g., PaLM-E {ref}, RT-2 {ref}, or LLaVA-based architectures {ref}, to generate interpretable, language-informed action sequences for the soft robotic platform. The planner will interpret high-level instructions such as “move the branch to the side, then harvest the ripe fruit,” and generate step-by-step commands conditioned on visual inputs and real-time scene understanding. The system will first construct a scene graph using outputs from RGB-D or multispectral cameras and semantic cues from VLMs. The LLM planner will then reason over object affordances (e.g., branch flexibility, fruit ripeness) and task context to generate actions for the robot’s arm and gripper. We will validate this framework in tomato and bell pepper harvesting, focusing on scenarios that require complex sequences, such as occlusion removal followed by precision picking.

Task 3.3.3: Natural Language Summarization of Multimodal Sensor Data for Farmer Decision Support

Beyond robotic autonomy, we aim to translate multimodal sensor outputs—visual, tactile, ethylene gas, and environmental measurements—into concise, uncertainty-aware natural language summaries for farmers. These summaries will support real-time human oversight and post-harvest analysis, addressing more open-ended questions like: “Which fruits were not harvested and why?” “Which regions of the greenhouse show signs of crop stress or delayed ripening?” “Which harvest events were uncertain or required replanning?” Using LLMs fine-tuned on structured sensor logs and annotated explanations, the system will generate queryable reports that identify anomalies, summarize system decisions, and highlight uncertainty or missing data. The outputs will be accessible via a voice or text-based interface and visualized in a dashboard for intuitive user interaction.

Expected Outcomes: This focus area will deliver a comprehensive perception and reasoning framework that enables soft robotic systems to interact intelligently with complex crop environments. By combining multimodal sensing, embedded AI models, and vision–language reasoning, we aim to improve the robot’s ability to detect, estimate, and respond to plant conditions in real time. These outcomes will enhance robotic decision-making, support early disease and stress detection, and generate actionable insights for growers in diverse field conditions.

- Multimodal sensing and fusion: A field-ready system that combines RGB-D, hyperspectral, thermal, and tactile data to map fruit, canopy structure, and stress indicators in real time.
- AI-enabled phenotyping and disease detection: Lightweight models for detecting fruit maturity and early-stage diseases, validated on specialty crops in both controlled and field environments.
- Crop estimation and digital twins: Digital plant models that link sensor data to yield and harvest predictions, supporting real-time decision-making and robotic task planning.
- Vision–language reasoning for robotics: A module that translates sensor outputs into natural-language summaries and task plans, guiding soft robot actions and supporting farmer interaction.

Focus Area 4: Decision-Making and General-Purpose Control in Soft Robotic Systems for Agile Harvesting

This research thrust will design planning and control frameworks that enable robots to execute complex agricultural tasks—such as harvesting, pruning, and pollination—under uncertainty and resource constraints. By developing algorithms that generalize across crops and tasks while leveraging the unique affordances of soft robots, including variable stiffness, distributed sensing, and safe contact. This thrust aims to tackle fundamental challenges in real-time control of high-DOF, compliant systems, hybridizing data-driven learning with model-based planning to achieve interpretable and robust behavior.

Objective 4.1: General-Purpose Agile Grasping and Manipulation using Soft Gripper

Scientific Rationale: Greenhouse harvesting demands high-speed, safe interaction with delicate and variable produce in cluttered environments. Traditional planning-control pipelines are either too slow due to complex dynamics or lack robustness under fruit variability. Occlusion, deformable body behavior, and inconsistent ripeness add additional complexity. To enable real-time, scalable deployment, we aim to build an integrated Real-to-Sim-to-Real (R2S2R) pipeline centered on simulation-grounded learning and residual real-world adaptation, with a core goal: How can we increase grasping speed without sacrificing accuracy or safety?

Research Plan and Methods: Develop general manipulation strategies using soft grippers and soft robotic arms developed in focus area 1 for agile and efficient picking of tomatoes and bell peppers, incorporating variable stiffness control and sensor feedback to handle fruit variability and minimize damage.

Task 4.1.1 Real2Sim Training

We propose a four-stage pipeline to close the real to sim-to-real loop for soft robotic manipulation in agricultural environments (developed in **Focus Area 1**), grounded in biological physical data and supported by online adaptation. In the first step that focus on **Real2Sim Deployment**, we begin by collecting physical data using RGB-D cameras to capture plant geometry and color, and by attaching wearable force/torque and displacement sensors to fruits such as tomatoes and bell peppers to measure physical interactions—such as force application and tissue deformation—during human-controlled grasps collected in **Focus Area 2 (Objective 2)**. Semantic segmentation models, including SAM and vision-language models (e.g., GPT 4.0), are employed to annotate the visual properties of the fruits (e.g., ripeness, occlusion). These data are used to generate semantic 3D point clouds annotated with both visual and mechanical properties. Next, we construct a soft manipulator simulation using MuJoCo (cite) and DisMech (cite), with additional modules like FruitGym (cite) and SofaGym (cite) to simulate fruit-stem-leaf assemblies with parameterized mass and stiffness derived from sensor measurements. These outputs form physics-aware assets that replicate real-world variability.

We then start Training in Simulation, we use reinforcement learning (e.g., PPO or SAC) with reward functions partially derived by language models. The reward structure encourages speed (minimizing time-to-grasp), safety (applying forces below damaging thresholds), and success (stable grasp and lift-off). Domain randomization is applied across physical properties (e.g., stiffness, friction) and visual attributes (e.g., lighting, fruit color), and curriculum learning gradually increases task complexity with challenges such as occlusion or fruit clustering.

Task 4.1.2 Sim2 Real Deployment

Next, we will begin **Sim2Real Deployment**, which involves deploying the trained policy on a real soft robotic arm, using RGB-D inputs and onboard force/torque sensing (without needing displacement sensors). The learned controller outputs control signals for trajectory tracking or hybrid impedance control strategies in real-world conditions. Finally, we start Online Adaptation, the system compares expected vs. actual grasp outcomes using feedback from onboard sensors and vision-based analysis of post-grasp deformation. Residual learning techniques update the policy in real time, enabling the robot to adapt its behavior based on discrepancies. If consistent failures are observed, a language model-based debugging agent is activated. This agent analyzes the task description, sensory logs, and control traces to generate a natural language explanation, propose revised reward functions, or recommend retraining. This stage ensures rapid policy refinement and minimizes downtime by incorporating human-in-the-loop feedback.

Objective 4.2: Imitation Learning for Robotic Manipulation in Indoor Environments

Scientific Rationale: Traditional programming and control methods often fall short in the complex, unstructured environments typical of small farms, where variability in crop types, plant structures, and terrain is harsh. Using imitation learning and demonstration-driven training to teach robots how to perform manipulation tasks in indoor high-tunnel and greenhouse settings, can enable quick adaptation to structured but dynamic agricultural layouts. By leveraging human expertise, imitation learning enables robotic systems to acquire complex manipulation skills without the need for exhaustive manual programming or large-scale labeled datasets. Through imitation learning, robotic manipulators can learn nuanced behaviors such as gentle fruit picking, selective harvesting, and obstacle avoidance, all while adapting to the unique characteristics of each farm. This approach not only accelerates deployment but also democratizes agricultural robotics, making advanced automation accessible to smallholder farmers.

Research Plan and Methods: We will develop architecture for teaching manipulator robots by demonstrations. The architecture will be mainly composed of state-of-the-art systems for teleoperation, data augmentation, and learning. To develop this architecture and achieve the proposed goals, this project has three main complementary tasks as described in the following subsections.

Task 4.2.1 Experimental Setup and Synthetic Data Creation

We will configure a setup consisting of an existing WVU manipulator robot, a gripper, a set of cameras, and a teleoperation device, all connected to a computer for data collection using the Robot Operating System (ROS2). To train the robot, we plan to use the Gello teleoperation device. This is a state-of-the-art, open-source framework for robot teleoperation composed of a scaled 3D-printed version of the robot being teleoperated and the software that controls the robot. The model is trained to generate future robot positions based on current joint positions and images for a specific task. The basic approach for generating training data is to perform the task by a human operator in a way that provides the system with the same type of inputs as it would experience when functioning on its own. Typically, this is done using a teleoperation setup [] where the operator controls a duplicate of the system to be trained (the leader), and

the movements are copied to the trainee system (the follower). In general, this type of training requires multiple demonstrations with varied inputs/configurations, and the performance of the resulting system is highly dependent on the quality of the demonstrations. Human-driven demonstrations, apart from being limited in the number of samples, can be suboptimal, inconsistent, or noisy. We plan to leverage synthetic data to reduce these issues.

The generated data will ultimately consist of targeted and observed robot joint angles and associated camera images. We plan to use Nvidia's GROOT-Cosmos pipeline [1], which features a two-stage approach to enriching the training data set. First, GROOT-Mimic up-samples a moderate set of trajectories obtained from human demonstrations using interpolation, while ensuring the physical consistency of the generated data. Next, the Cosmos Foundation Model [2] takes these synthetic trajectories as inputs and generates corresponding videos. This generation pipeline is conditioned on the images corresponding to the original trajectories derived from the human demonstrations. The entire data augmentation process results in an exponential increase in the number of trajectory and consistent videos while at the same time making sure that there are no violations of basic physics in the generated paths.

Task 4.2.2 Teaching by Demonstration Algorithm

The main idea behind teaching robots by demonstrations is to learn a closed-loop policy that represents the underlying dynamical system of the robotic setup. The policy aims to predict a sequence of actions given a sequence of observations. A policy can be created with several tools including Gaussian Mixture Models [3], System Identification, as shown in Dr. Pereira's previous work [4], and, more recently by generative models (GMs) [5]. The main advantage of the new approaches based on GMs is that end-to-end solutions can be created, with no need for data preprocessing. These approaches can learn a policy that directly maps sensor data (e.g., images) to actions (e.g., robot joint angles).

For the core algorithm, in this project, we will build on the Action Chunking with Transformers (ACT) approach proposed by [6]. This formulation has been successfully implemented on a low-cost platform. One salient element is decomposing the action into short sequences of steps ("chunks"), and training models to predict the next chunk of steps based on the feedback received in the current chunk. The prediction mechanism relies on transformers [7]. During training, the leader's joint angles are recorded as "actions". The joint angles of the follower and corresponding video streams constitute the "observations". For a given task, the ACT algorithm trains an instance of the CVAE (Conditional Variational Autoencoder) model to predict the actions for a sequence of k time steps from observations during the preceding sequence of k steps. During execution (deployment), the robot feeds its observed position (joint angles) and its video to a (pre-selected) trained instance of the CVAE, which in turn provides target angles for the next time steps.

Preliminary Results: PI Pereira is currently working on how to teach robots on pepper picking with WVU REU students. An illustration of our laboratory setup is shown in Fig. ZZ. Initial results indicate that more training data is necessary to increase the success rate of the robot on the task execution, what motivates the previous methodology for data augmentation. The state-of-the-art system for this kind of task reports a success rate of only 28.95% with a cycle time of 31.71 seconds when the robot operates in outdoor crops, which may not be acceptable for practical use [8] and motivates the realization of this research.

Objective 4.3: Mobile Manipulation for Greenhouse and Outdoor Harvesting

Scientific Rationale: Once manipulators are developed and trained to harvest when they are in front of a plant, they need to be deployed in the greenhouse or the outdoor crop. It is our goal to integrate one or more manipulators into wheels- or whegs-based mobile platforms capable of navigating narrow rows of uneven terrain and recognizing target crops. Once in front of a plant, the mobile-manipulator system will leverage the training performed in the previous objectives to perform the harvesting.

Research Plan and Methods:

Task 4.3.1 Hardware Setup

We will integrate our manipulators first, into a commercial mobile platform, such as the Husky mobile base provided by ClearPath robotics, and second into a WVU URC base equipped with whegs and those developed in **Focus Area 2**. To interact with its environment, the robot will also be equipped with Global Positioning System (GPS), Inertial Measurement Unit (IMU), light detection and ranging sensor (LIDAR) and a RGB-D camera. The sensors will be used for obstacle avoidance and robot localization. The robot components and the sensors will be controlled by a single computer, which will be housed inside the robot base.

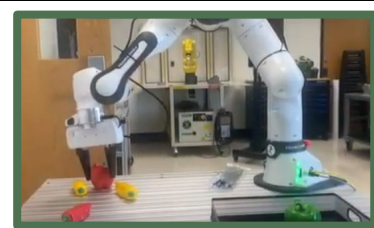


Fig. 1: Laboratory setup for teaching robots by demonstration.

Task 4.3.2 Software Development

We will create easy to test and evaluate building blocks that can be combined to create complex robot behaviors. The robot software will be based on the ROS2 to allow the use of advanced public toolboxes made freely available on-line. It is our expectation that, at the conclusion of this objective, we will have a collaborative robot that can execute specific commands given by the farmer, which includes at least “follow me”, “go home”, and “start harvesting”. These commands will be given by gestures, voices or by pressing a single button, and may include a direct interaction between the farmer and the robot, for example, to move the robot manipulator to a desired position. These simple commands can be used by the farmer to move the robot to the harvesting place and start the harvesting task.

Research Capacity Building

SOFT-AG will establish several advanced robotic manufacturing capabilities to support the development of multifunctional soft robotic systems. For example, our lab is equipped with a Carbon 3D printer, enabling multi-material fabrication of soft robots and actuators using both rigid and elastomeric photopolymers. In addition, a micro flexible PCB fabrication facility to support the integration of compact, lightweight electronics directly into soft robotic substrates will be developed. We are also building a traditional extrusion-based printing system specifically designed for hydrogel-based optoelectronic sensors. It will contribute new techniques in real-time learning and estimation for deformable coupled systems and advance the broader understanding of compliant manipulation in occluded, variable, and contact-rich settings.

D. Workforce Development and Inclusive Ag-Tech Innovation

Workforce development efforts for this project will leverage several successful robotics and STEM education programs already implemented in WV. This approach will allow for the creation of new activities and lesson plans centered around soft robotics and agricultural robotics, while minimizing the development time and effort and accelerating the impact and timeline. The NASA IV&V Facility Education Resource Center (ERC), which has been providing K-16 STEM and robotics outreach for 25+ years, will work closely with team to ensure engagement.

Several existing initiatives that will be leveraged focus on youth education and utilize partnerships among classroom teachers, higher education institutions, and university students. The NASA ERC Loan and Learn Program provides in-person workshops and online training for formal teachers and informal educators in the use of STEM and robotics kits, which can be borrowed for classroom use after the completion of training. Both NASA ERC and WVU offer STEM and robotics-themed summer camps for K-12 youth, which are staffed in part by undergraduate and graduate students. In addition, WVU Statler College offers hands-on STEM activities for school through field trips and classroom visits, working with **dozens of schools per year**. The WV Science Public Outreach Team (WVSPOT) trains undergraduates (including at WVU and MU) to deliver interactive presentations to K-12 students about STEM topics and careers. All these programs will be excellent platforms for sharing soft robotics and robotic agriculture with youth across the state.

Robotics education extends beyond the classroom in WV, offering further partnerships for this project. The WV Robotics Alliance organizes robotics competitions for elementary school through college. The 500+ competitive robotics teams across the state will be a natural audience for outreach tied to this project.



Fig. 2: SOFT-AG Execution Plan Through Phase 1.

WVU's 4-H STEM Ambassadors program trains university students to teach STEM curriculum at 4-H camps and community events every summer. With a presence in every county in the state and a long history of both STEM and agricultural programs, 4-H is another strong partner for this project. Specific outreach activities in soft robotics and robotic agriculture will be developed in conjunction with PIs and be directly inspired by their research. For example, LEGO and VEX Robotics teams could compete to "harvest crops" using robots, high school students could design and create soft parts for a robotic arm, or middle schoolers could learn to cope with color sensors to distinguish between ripe and unripe strawberries.

E. Jurisdictional Impact and Sustainability

The SOFT-AG team has an established track record on competitively funded robotics and robotics agriculture research from several federal agencies; however, this work will be transformative in aligning our entire jurisdiction in the highly interdisciplinary field of soft robotics for agriculture applications. WVU has recently invested in several new faculty lines in robotics (e.g., grown from 2 to 5 tenure track faculty in the past 5 years) and has developed a new engineering B.S. degree program to continue to grow this pipeline of students and faculty lines. To support lower teaching loads for research focused faculty, WVU engineering has also grown the number of teaching focused faculties in the college. We will leverage and build upon our existing programs such as the state funded STEM SURE program and our NSF REU which focuses on robotics in rural Appalachia with emphasis on students from the Appalachian region to sustain research networks and collaborations across the institutions. Under this award, we will initiate a regional workshop focused on robotics for Appalachia with agriculture being a central theme. We will leverage our jurisdictions proximity to major play. We will also seek support through the WV Research Challenge fund, a program referenced in the WV Vision 2024 S&T plan to sustain and grow collaborations.

II. Execution Plan, Evaluation, and Assessment

A. Execution Plan

The team will maintain a collaborative environment through quarterly meetings (two virtual and two in-person) across institutions, monthly meetings across focus areas, and an annual workshop that engages all project stakeholders including the NSF program officers, the project personnel, graduate students, evaluation team, and our external advisory board. Each focus area will meet monthly with report outs to the overall project team at quarterly meetings. As detailed in the proposal, research thrusts are broken down to the task level. Co-PIs will lead each of the 4 thrusts, with members of the team leading each individual task as shown in Fig. AA

Table 1 Exal Stuff

Evaluation Activity	Y1	Y2	Y3	Y4
Refine evaluation plan	X	X	X	X
Develop & refine evaluation instruments	X	X	X	X
Administer annual progress survey, includes formative questions	X	X	X	X
Administer formative activity surveys	X	X	X	X
Interviews with project participants		X	X	X
Formative evaluation updates	X	X	X	X
Annual evaluation report	X	X	X	
Summative evaluation report				X

B. Evaluation and Assessment Plan (1.33 page NOTE TO REVIEW, this is a placeholder EVAL section)

Project evaluation will be led by Lynnette Michaluk PhD, WVU Center for Excellence in STEM Education. She is research mentor on two BCSEI-IIDs (2126018, PI McEldowney, 2022-2025; #2320120, PI Zhang, [recently terminated, work continues unfunded]) and is PI on NSF CS for All Secure and Upgrade Computer Science in Classrooms through an Ecosystem with Scalability & Sustainability (SUCCESS, #2031355, 2021-2026). She

serves as evaluator on several funded projects including NSF RII Track 2 FEC: Enabling Factory to Factory (F2F) Networking for Future Manufacturing (OIA-2119564), Department of Education Title III Strengthening Potomac State College, and NASA Plant the Moon Challenge (#80NSSC22M0308). She has authored/co-authored several items in STEM education (see biosketch). Two research associates will assist Michaluk. Evaluation will include both summative and formative evaluations. Summative evaluation will focus on impact and assess progress made toward achievement of project goals and the broader impacts on students, faculty, participating institutions, and communities. Guiding evaluation questions are aligned with the goals of the project. Table 3 outlines the timeline of evaluation activities.

Table X. Evaluation Metrics, Concepts Measured, Assessment Methods and Data Collection Timeline

Concepts Measured	Assessment Method	Data Collection Timeline	Comparison (Analysis)
1) Factory to Factory Network: To what extent has the project developed the foundation for a network integrating manufacturing with artificial intelligence?			
Achieved expected outcomes of research objectives related to pillars 1 – 5	Interviews with key participants;	End of each project year	Compare themes between time points
2) Education & Workforce Development: To what extent has the project developed a sustainable and inclusive workforce in high-tech areas of FM via the proposed LLP-FM?			
- Increased middle and high school student participants' awareness of and readiness for FM; - Development of FM mini-lessons, PD materials, micro-credentials, and online program	STEM-WELM model-based survey; Surveys, documents, lessons, related artifacts	At program entry (baseline) & end of each academic year	Baseline-post (paired samples <i>t</i> -test with confidence interval and effect size); Document analysis
- Engagement with K-12 schools (# of schools reached, #students reached via visits & activities, # of teachers reached) # Undergraduate students and graduate students entering internships and workforce	Program enrollment tracking data (numeric)	End of each academic year	Actual to target (descriptive)
(3) Collaboration & Outreach: To what extent has the project fostered collaborations on FM, LLP-FM and F2F networking in WV and SC, and between community members and other researchers?			
- Increased cross-disciplinary collaboration (joint publications, tutorials, joint projects, joint funding, etc.) - Increased interaction between team members (community & industrial partners, joint mentoring of new faculty, joint student supervision, visits and seminars)	Baseline and annual survey; Interviews with program administrators, faculty, industrial partners	At program entry (baseline) & end of each academic year;	Baseline-post (paired samples <i>t</i> -test with confidence interval and effect size); Compare themes between time points

Table X provides evaluation questions, project concepts to be measured, assessment methods, data collection timeline and comparisons to be conducted, with a brief analysis description. Formative evaluation monitors quality of project activities and provides feedback to the leadership team to improve project implementation, maximize impacts, and document/share lessons learned. Formative evaluation questions include: (1) Is the project team implementing effective strategies and activities to achieve project goals? (2) Is the project implementation on schedule and expected to meet output targets? (3) What are the supports and barriers to project implementation? (4) Have unexpected outcomes occurred because of implementation, and if so, how have they been addressed by the team? To address these questions, the evaluator will: (1) Conduct interviews with key project faculty to assess supports and barriers to project implementation; (1) Survey student and teacher participants about their experiences in hands-on and online training; (3) Conduct interviews with industrial partners faculty, graduate students, and undergraduate students about their experiences with program activities; (4) Track outputs relative to targets. The summative evaluation report will demonstrate the project's impact. As part of the evaluation and data management plans, project leads will share evaluation and other data with EPSCoR officers for centralized project data collection and will provide all required data in a timely manner.

III. Organization and Management

A. Management Plan and Institutional Contributions

To ensure a thriving ecosystem withing SOFT-AG, the research team comprises a total of four research universities across the state jurisdiction including: WVU (lead), Marshall University, West Virginia State University, and Fairmont State University, with research expertise in robotics (5 faculty), control and control (2 faculty), hydrology (1 faculty), biology (1 faculty), and K-12 STEM outreach through competitive robotics (1 faculty). Dr. Jason Gross (PI, WVU), Professor and Chair of the Dept. of Mechanical, Materials and Aerospace Engineering (MMAE) will serve as the overall administrative lead for the project, ensure project reporting, and regular meetings of the team and internal and external stakeholders. Dr. Anand Mishra (Deputy-PI) is an Assistant Professor in Robotics in the WVU MMAE department and will serve as research lead and lead Focus Area 1 with the support of robotics professor Dr. Yu Gu (SP, WVU). Dr. Jason Hubbart (Co-PI, WVU) is a Professor and Assoc. Dean for Research in the WVU Division of Land Grant engagement

will lead Focus Area 2 with the support of Dr. Nimmakayala (SP, WVSU), Biology Professor. Focus Area 3 will be led by Dr. Zhu (Co-PI, Marshall) and Dr Sardahi (SP, Marshall) who brings expertise in multi-agent estimation and control. Robotics faculty Dr. Pereira (Co-PI, WVU) and Dr. Szorcinski (SP, WVU) will lead focus area 4. Finally, Dr. Ensign (SP, FSU) will lead educational outreach for the project with the support of Ms. Harmon (SP, WVU), program assistant for robotics outreach at WVU.

Dr. Mishra is a new WVU hire who brings expertise in soft robotics for agriculture and will play a lead role on the science team with plans to survey as PI of the 3-year expansion project. As such, this project is in direct support faculty development toward capacity building in soft robotics for agriculture

The SOFT-AG advisory board will include some of the most prominent academic researchers in agriculture and robotics for agriculture and experts in robotics technology transfer. These include Dr. Michael Gore, Professor of Plant Breeding at Cornell, Dr. George Cantor, Agriculture Robotist at CMU's Robotics Institute, Dr. Girish Krishnan, Soft Robotics Expert at University of Illinois, Dr. Abraham Stroock, Professor of Chemical and Biomolecular Engineering at Cornell, and Dr. Yu Jiang, Assistant Professor of Plant Science, Cornell. Mr. Jeff Legault, Associate Director of CMU's National Robotics Engineering Center will also participate to provide insights on technology commercialization. Members internal to the jurisdiction. Lewis Jett, WVU Extension Specialize for Commercial Horticulture, and Mr. Timothy Winfield, Farm Supervisor at WV's USDA Kearnesville Tree Fruit Farm. The board along with the evaluation team and NSF POs will convene at the project kickoff annual project meetings in support of continuous improvement.

IV. Results from Relevant Prior Support

Gross, Pereira: (a) NSF # 1851815, 04/2024 – 03/2027, 454,179. (b) REU Site: Undergraduate Robotics Research for Rural Appalachia (c) Intellectual Merit: research in the core robotics areas of motion planning, localization, coordination, autonomy, and locomotion. Projects will aim for demonstration in challenging field applications in rural settings. (d) Broader Impacts: the REU site creates research and learning opportunities for students primarily from the Appalachian region (e) Products: []

Gu, Gross (a) NSF # 2132937, 08/2022 – 08/2025, \$366,796. (b) Collaborative Research: NRI: Reducing Falling Risk in Robot-Assisted Retail Environments (c) Intellectual Merit: investigating the health and safety implications of the wholesale and retail crew collaboratively working with robots in a typical retail environment. (d) Broader Impacts: Reducing potential of slip and fall in retail space; providing educational and training opportunities to diverse groups (e) Publications: []

Gu, Gross (Waterland and Hu) also have a USDA NRI StickBug. Intellectual Merit: Develop a novel precision pollinator robot that work alongside humans and demonstrate technologies for artificial robotic pollination of brambles and tomatoes; Broader Impacts: develop robot technologies to enhance agriculture in light of declining honeybee populations Products

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